

Amith Epuri

Sr. Data Engineer

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PROFESSIONAL SUMMARY:

- **10+ years of experience** as a Data Engineer delivering **cloud-native data solutions** across **AWS, GCP, and Azure**, including **lift-and-shift migrations, real-time/batch ETL pipelines, data integration, quality validation, and automation** using **Python, Airflow, and Databricks**.
- Extensive experience deploying and managing **AWS cloud applications**, leveraging **EC2, S3, Glue, Lambda, EMR, Redshift, RDS, DynamoDB, Kinesis, Step Functions, CloudWatch, IAM, Auto Scaling, and Athena** for scalable, high-performance data solutions.
- Designed and implemented **ETL migration services using AWS Lambda and Glue Catalog**, building **serverless pipelines** that automate ingestion, transformation, and analytics in **Athena and Redshift**.
- Built and optimized **data lakes on S3**, implementing **partitioning, lifecycle policies, and Iceberg-based schema evolution** to support time-travel queries and cost-effective historical data management.
- Developed and orchestrated **data pipelines on AWS EMR** using **Spark, PySpark, and Hive**, optimizing performance through right-sizing, auto-scaling, and spot instance utilization for cost efficiency.
- Designed **real-time streaming and event-driven architectures** using **Kinesis, Kafka, and Lambda**, enabling near real-time data ingestion and transformation for analytics and ML workloads.
- Architected and maintained **data warehousing solutions in Snowflake and Redshift**, developing ELT transformations, implementing **data quality checks**, and enabling cross-cloud analytics between **AWS and GCP**.
- **Implemented cost optimization strategies** in AWS by right-sizing EMR clusters, leveraging Spot instances, and automating lifecycle policies on S3 buckets — reducing overall cloud spend by ~30%.
- **Integrated AWS Lake Formation** for centralized data governance and fine-grained access control, ensuring secure data sharing and compliance across analytics teams.
- **Deployed scalable analytical environments using Amazon Redshift Spectrum and Athena**, enabling federated queries across S3 and Snowflake without data movement.
- **Collaborated with cross-functional teams** to modernize on-prem data ecosystems by migrating ETL workloads to **AWS Glue and EMR**, improving job performance and maintainability through modular Spark codebases and Airflow orchestration.
- Implemented **Infrastructure-as-Code (IaC)** using **Terraform, Terraform Enterprise, and HCP** to provision AWS infrastructure, and integrated **Concourse CI and GitHub Actions** for continuous delivery of data pipelines.
- Configured **AWS CloudWatch, SNS, and CloudTrail** for real-time monitoring, logging, and alerting, ensuring pipeline observability, reliability, and compliance across distributed systems.
- Delivered **end-to-end Big Data solutions** migrating legacy ETLs to **AWS Glue, EMR, and Step Functions**, enabling high-performance data transformations and reduced operational overhead.
- Built **data integration pipelines** between **S3, Snowflake, APIs, and streaming sources**, supporting both batch and real-time processing for analytics and business intelligence workloads.
- Developed **Python-based ETL scripts** using **Pandas, NumPy, and Boto3**, automating data ingestion, cleansing, and transformation across multiple AWS services.
- Implemented **CI/CD pipelines** using **GitLab and AWS CodePipeline**, automating data infrastructure deployment, testing, and monitoring for improved agility and governance.
- Designed and maintained **Power BI and Tableau dashboards** on Snowflake and Redshift datasets, optimizing performance and scalability for executive-level reporting.
- Hands-on experience with **Hadoop ecosystem components**—HDFS, Hive, Spark, Kafka, Sqoop, YARN—and their **integration with AWS-native data services** for hybrid analytical workflows.
- Strong in **SQL and NoSQL databases** (MySQL, Oracle, DynamoDB, MongoDB, HBase) and **data modeling** using RDBMS concepts like views, stored procedures, and triggers.
- **Collaborated with cross-functional data, DevOps, and analytics teams** to implement end-to-end AWS data architectures using **Glue, Lambda, EMR, and Redshift**, ensuring scalability, governance, and alignment with enterprise data strategy.

TECHNICAL SKILLS:

Bigdata Ecosystems	HDFS, YARN, MapReduce, Sqoop, Hive, Oozie, Pig, Spark, Zookeeper, Cloudera Manager, Flume, NiFi, StreamSets
Hadoop Distributions	Apache Hadoop, Cloudera CDP, Hortonworks HDP
Cloud Technologies and Services	Azure: ADLS/Blob, Azure Data Factory (ADF), Azure Databricks (ADB), Synapse Analytics, Key Vault, Logic Apps, Azure Functions, Service Bus, Event Hubs, Cosmos DB AWS: S3, Amazon Glue, Kinesis, EMR, Step Functions, Lambda Functions, IAM GCP: GCS, DataProc, BigQuery, Pub/Sub, Dataflow, Cloud Composer, Lambda Architecture
Scripting Language	PythonPython, Java, Scala, R, Shell Scripting, HiveQL
NoSQL Database	Cassandra, MongoDB, HBase
Database	MySQL, Oracle, Teradata, MS SQL Server, PostgreSQL, DB2, Snowflake, Amazon Redshift, Azure Synapse Analytics
BI & Visualization	Power BI, Tableau
Web Development	JavaScript, NodeJS, HTML, CSS, Spring, J2EE, JDBC, Okta, Postman, Angular, Mockito, Flask, Hibernate, Maven, Tomcat, WebSphere
Version Control	Git, SVN

PROFESSIONAL EXPERIENCE:

Caresource, Dallas TX
Sr.Data Engineer

Mar 2024 to Present

Responsibilities:

- Spearheaded a team of data engineers, providing **mentorship, guidance, and technical leadership** to drive successful **project deliveries**.
- Developed and optimized **ETL pipelines using AWS Glue, PySpark, and EMR**, efficiently processing structured and semi-structured data (JSON, ORC, Parquet) stored in **Amazon S3**.
- Built **real-time streaming pipelines** leveraging **Amazon Kinesis Data Streams, Kafka, and Apache Flink**, applying complex business transformations for high-throughput, low-latency processing.
- Designed and deployed **containerized workloads using Docker and Amazon EKS (Kubernetes)** for high availability, scalability, and cost optimization across distributed data pipelines.
- Transformed legacy SQL logic and DBT models into **AWS Glue jobs orchestrated via Step Functions and Airflow**, optimizing performance and reducing cloud licensing overhead.
- Engineered **batch and streaming ETL workflows** using **AWS Data Pipeline, Lambda, and SQS**, enabling event-driven data ingestion and transformation for downstream analytics.
- **Implemented Infrastructure-as-Code (IaC) and CI/CD automation** using **Terraform, Terraform Enterprise, and HashiCorp Cloud Platform (HCP)** to provision multi-cloud resources; integrated **Concourse CI** and **GitHub Actions** for continuous deployment of data pipelines, ensuring reproducible, scalable, and secure infrastructure delivery across AWS
- **Migrated on-premises and legacy systems to AWS Snowflake and Redshift**, improving scalability, query performance, and reducing infrastructure management costs.
- Successfully led the migration of on-premises or legacy data systems to **Snowflake**, ensuring a smooth transition and improved data accessibility.
- Successfully led the migration of on-premises or legacy data systems to **Snowflake**, ensuring improved scalability, faster query performance, and enhanced analytics capabilities.

- Designed **Snowflake and Redshift integration pipelines** to ingest data from S3, APIs, and real-time streams, enabling unified enterprise data warehousing and analytics.
- **Deployed and managed AWS Control Tower** to establish a secure, governed multi-account landing zone for CareSource's cloud ecosystem, standardizing resource provisioning and access control across environments.
- **Automated enterprise account creation and configuration** using **AWS Organizations, Service Catalog, and CloudFormation StackSets**, ensuring compliance with internal guardrails and SOC 2 security policies.
- **Integrated Control Tower with GuardDuty, CloudTrail, Config, and Security Hub**, implementing centralized logging, real-time compliance alerts, and automated remediation workflows for continuous governance.
- **Collaborated with Cloud Ops and Security teams** to design baseline blueprints for networking, IAM, and encryption policies, improving consistency and audit readiness across 20+ AWS accounts.
- Implemented data quality checks and validation processes in **Snowflake** to maintain data accuracy and consistency.
- Created comprehensive **UAT** test plans, defining test cases, scenarios, and expected outcomes in collaboration with stakeholders to ensure alignment with project requirements.

Environment: AWS S3, Glue, EMR, Redshift, Lambda, ECS, EKS, CloudFormation, Step Functions, Snowflake, Databricks, Spark, PySpark, Spark-SQL, Spark-Streaming, Kafka, Airflow, Terraform, Hive, HDFS, Scala, Python, SQL Server, T-SQL, MongoDB, HBase, Docker, Kubernetes, Apache NiFi, Impala, Zookeeper, Sqoop.

Simmons Bank, Pine Bluff AR
Sr. Data Engineer

July 2022 to Mar 2024

Responsibilities:

- **Involved in the end-to-end Big Data flow**—from data ingestion from upstream systems to HDFS, through distributed processing, cleansing, and analytics using Spark and Hive.
- Extracted and analyzed data from **HDFS using Hive and Spark (Scala/PySpark)**; leveraged **Amazon Redshift** for advanced analytics, feature selection, and model preparation.
- As part of **MLOps modernization**, migrated multiple PySpark-based ML models from on-prem Hadoop to **GCP Dataproc and Oozie**, ensuring scalability, reproducibility, and reduced maintenance overhead.
- Designed **Airflow scheduling and orchestration** to optimize DAG execution times and resource utilization across **AWS and GCP pipelines**.
- Leveraged **GCP Dataflow's stateful processing and Apache Beam windowing** for real-time event-stream analysis with event-time and processing-time semantics.
- Developed Spark APIs to ingest data from **Teradata and Oracle into HDFS**, creating Hive schemas for analytical workloads.
- Extensively implemented **Terraform and CloudFormation** for provisioning AWS and GCP infrastructure, including projects, Kubernetes clusters, IAM roles, and networking resources.
- Built **fault-tolerant, retry-enabled pipelines** with robust error handling, ensuring high availability across multi-cloud environments.
- Designed and managed **data ingestion pipelines integrating BigQuery, S3, Google Cloud Storage, and third-party APIs**, enabling unified data lakes and downstream analytics.
- Built and optimized **Snowflake data warehouses** for analytics and BI dashboards, using partitioning, clustering, materialized views, and time-travel features.
- Developed **Snowflake SQL and Python scripts** to transform raw data into curated analytical datasets supporting self-service Power BI and Tableau dashboards.
- Created **real-time dashboards** visualizing machine-learning predictions and KPI metrics for immediate business insights using AWS QuickSight and Google Data Studio.

- Managed **Druid clusters** for streaming analytics—monitoring health, backups, and scaling strategies for low-latency query performance.
- Implemented **AWS Glue ETL pipelines** for serverless data integration and transformation into data lakes and Snowflake/Redshift warehouses.
- Developed **Hive and Sqoop jobs** to import data from Oracle/Teradata into HDFS (Avro format), and created partitioned/bucketed Hive tables in Parquet with Snappy compression.
- Optimized **AWS EMR costs** via auto-scaling and spot instances, leveraging **Apache Iceberg** for efficient historical data management, time-travel queries, and schema evolution.
- Configured **auto-scaling and monitoring on AWS and GCP**, enabling elastic compute allocation per workload for cost efficiency and performance.
- Implemented **CI/CD pipelines** for data engineering workflows using GitLab, Jenkins, and Cloud Build, automating testing, deployment, and monitoring across multi-cloud environments.
- Built **AWS Lambda and Kinesis pipelines** to automate real-time data ingestion and processing, reducing latency and enhancing scalability for streaming applications.
- Designed **Flume ETL jobs** to ingest HTTP data sources into HDFS for further processing in Spark and Hive.
- As part of **MLOps**, worked on the migration of various **PySpark models** from on-prem Hadoop clusters to **GCP** using **Dataproc** and **Oozie** workflow.
- Implemented **scheduling strategies** within **Airflow** to optimize job execution times and resource utilization.
- Leveraged **GCP Dataflow's** stateful processing capabilities to maintain and analyze stateful information across data processing windows, enhancing insights from streaming data.
- Implemented **windowing strategies** in **Apache Beam** for time-based data processing, efficiently handling event-time and processing-time semantics.
- Extensively implemented **Terraform** to create **GCP projects, Kubernetes clusters**, and other GCP resources.
- Developed **data ingestion pipelines**, integrating **BigQuery** with various data sources including **Google Cloud Storage**, Google Sheets, and third-party platforms.
- Designed **AWS Lambda-based data pipelines** to automate ingestion and processing workflows, reducing latency and enhancing scalability. Designed data pipelines using **AWS Lambda** functions to automate data ingestion and processing workflows, reducing latency and enhancing scalability.

Environment: HDFS, Yarn, MapReduce, Hive, Sqoop, Flume, Oozie, HBase, Kafka, Impala, Spark SQL, Spark Streaming, Eclipse, Oracle, Teradata, PL/SQL, UNIX Shell Scripting, Cloudera, Cloud Dataflow, AWS Lambda, Terraform, GCP, GKE.

Ulta Beauty, Bolingbrook IL
Data Engineer

Apr 2020 to June 2022

Responsibilities:

- Worked with **Azure cloud services** including **HDInsight, Databricks, Data Lake, Blob Storage, Data Factory, and Storage Explorer** for scalable data processing and storage.
- Analyzed data from **Azure Data Lake** using **Databricks** and **Spark clusters** to derive actionable business insights.
- Implemented robust **monitoring and alerting solutions for Airflow workflows**, proactively addressing potential issues and minimizing downtime.
- Established **KPIs** to measure and report on data pipeline reliability and performance.
- Designed and developed **Power BI dashboards** to track KPIs, sales trends, and ML model predictions, providing stakeholders with real-time insights.
- Integrated **Power BI** with **Snowflake** and **Azure Synapse** to deliver unified reporting across multiple business units.
- Developed **data ingestion, aggregation, and integration pipelines** using **Snowflake** and **Azure**

Data Factory for advanced analytics.

- Created and managed **ADF pipelines and linked services** to extract and transform data from diverse sources.
- Developed **Spark/Scala scripts** for data extraction, transformation, and analysis, leveraging **partitioning, caching, and checkpointing** for optimized performance.
- Implemented **real-time data processing** using **Spark Streaming and Kafka**, transforming data into DataFrames and loading results into **HBase**.
- Utilized **Terraform** to automate provisioning and management of cloud infrastructure, including Kubernetes clusters, networking, and storage.
- Worked with **AWS services (S3, Glue, Lambda, Redshift, Athena, Spectrum)** to enhance data integration, processing, and analytics workflows.
- Developed **MapReduce jobs in Python** for distributed data processing and implemented **automation scripts** for ETL validation across databases like Oracle, SQL Server, and Hive.
- Used **Azure DevOps** for **CI/CD** and production deployment, ensuring smooth delivery of data engineering solutions.
- Leveraged **Azure Databricks** for **collaborative data engineering**, implementing ML models and optimizing Spark jobs for efficient processing.
- Designed **Power BI dashboards** and performed **issue resolution and monitoring** of production pipelines.

Environment: HDFS, Yarn, MapReduce, Hive, Sqoop, Flume, Oozie, HBase, Kafka, Impala, Spark SQL, Spark Streaming, Eclipse, Oracle, Teradata, PL/SQL, UNIX Shell Scripting, Cloudera, Cloud Dataflow, AWS Lambda, Terraform, GKE, Python, PySpark, SQL, GCP.

Consumer Energy, MI
Data Engineer

Jan 2019 to Mar 2020

Responsibilities:

- Installed and configured **Apache Hadoop** to test and maintain log file management across the Hadoop cluster.
- Installed and configured **Hive, Pig, Sqoop, and Oozie** on the Hadoop cluster to facilitate data integration, ETL processes, and workflow automation for optimized data processing.
- Proficient in **SQL** programming, with hands-on experience in database design, query optimization, and performance tuning across **MySQL** and **SQL Server** for enhanced data management and retrieval.
- Led ETL migration from **Oracle** to Hive, developing Hive queries, **UDFs**, and optimizing Pig scripts to improve performance and reporting accuracy.
- Utilized **Sqoop** for efficient data ingestion into **HDFS** and Hive, developed **MapReduce** jobs, and orchestrated automated workflows using **Oozie** for seamless data processing pipelines.
- Developed **Spark** scripts using **Scala** to perform distributed data processing and improve overall throughput on the Hadoop ecosystem.
- Built scalable distributed data solutions using Hadoop, ensuring high availability and fault tolerance across data pipelines.
- Hands-on experience in **Python** scripting for data manipulation, automation, and orchestration tasks within ETL workflows.
- Worked with **NoSQL databases** such as **HBase, Cassandra, and MongoDB** for data warehousing and mining applications.
- Monitored and managed Hadoop clusters using **Cloudera Manager**, ensuring performance optimization and cluster stability.
- Implemented **CI/CD pipelines** for data engineering workflows using **Jenkins, GitLab**, and **AWS CodePipeline**, enabling automated deployment, testing, and monitoring.
- Developed and deployed **AWS EC2** instances using **Terraform** and Python, integrating **AWS EFS** to enhance job processing throughput and reduce operational costs.
- Utilized **AWS Glue** for ETL processes, integrating and transforming data from multiple sources, and

optimized **Amazon Redshift** for large-scale analytical queries.

- Implemented **AWS Lambda** functions for serverless data processing and developed real-time analytics pipelines integrating data from **Amazon RDS** and other sources.
- Managed infrastructure deployment using **AWS CloudFormation**, and optimized resource utilization with **AWS Batch**, **Amazon EMR**, and **AWS Kinesis** for large-scale and streaming data workloads.
- Ensured data integrity, security, and cost efficiency by implementing **AWS IAM**, **S3 lifecycle policies**, **S3 Glacier** archival storage, and **AWS Backup** for disaster recovery and compliance.
- Managed data ingestion and processing using **Azure Data Lake**, **Azure Storage**, **Azure SQL**, and **Azure Databricks** for scalable analytics.
- Developed and maintained tabular models in **Azure Analysis Services** to improve business reporting and enable faster access to analytical insights.
- Implemented **Azure Data Factory** pipelines to orchestrate data workflows, automate data movement and transformation, and ensure end-to-end ETL scalability across multiple sources.

Environment: Apache Hadoop, Cloudera Manager, CDH2, Python, CentOS, MapReduce, Azure, Pig, Hive, Sqoop, Oozie, SQL, AWS Glue, DynamoDB, AWS Lambda, S3, Redshift, CloudFormation, Kinesis, EMR, AWS Batch, Terraform, EC2, Jenkins, GitLab, AWS CodePipeline.

Flipkart, INDIA
Data Engineer

June 2015 to Apr 2018

Responsibilities:

- Developed ETL pipelines using Hive (HQL) and SQL (Oracle) to extract, transform, and load data for analytical reporting.
- Performed data wrangling, cleaning, and transformation on structured and unstructured datasets using NumPy, Pandas, and PySpark, ensuring high data quality.
- Optimized SQL queries and PL/SQL procedures through execution plan analysis to enhance database and query performance.
- Created and managed Hive, Phoenix, and HBase tables using ORC format and Snappy compression for efficient storage and faster processing.
- Automated incremental data loads and end-to-end data workflows using Oozie, improving data pipeline reliability and performance.
- Utilized Sqoop to import data from Oracle to Hive, converting it into Parquet format for optimized querying and analytics.
- Leveraged Apache Spark on Databricks for distributed data processing, enabling scalable and real-time analytics.
- Implemented Teradata utilities (Fast Export, MLOAD) for large-scale ETL and data migration between OLTP and OLAP systems.
- Automated serverless data processing using AWS Lambda, integrating with S3 and RDS to streamline workflows and reduce operational costs.
- Designed and deployed predictive analytics and machine learning models in Databricks, visualizing insights through Tableau dashboards hosted on AWS (S3/EC2).
- Managed AWS EC2 instances for scalable execution of Spark jobs and machine learning models.

Environment: Python, ETL, Agile, Data Quality, Tableau, Data Governance, Supervised & Unsupervised Learning, NumPy, SciPy, Hadoop, Sqoop, HDFS, Spark SQL, Pandas, PostgreSQL, AWS (EC2, RDS, S3), Matplotlib, Scikit-learn, Shiny.